

Prompt Engineering Iteration Log

Objective

The objective of this exercise was to improve a prompt one technique at a time using a real task from my FL-01 workflow audit. After each iteration, I compared the output with the previous version to understand how each prompt engineering technique influenced the quality of the response.

Selected Task

Task: Write a portfolio-ready case study about my FlyRank AI Engineering Internship for recruiters seeking AI/ML internship or entry-level opportunities.

Version 0 : Naive Prompt

Technique

None (Naive Prompt)

Prompt

Write a case study about my internship.

Representative Output

During my internship, I gained valuable experience working on real-world projects and developing my technical skills. I collaborated with team members, learned new technologies, and improved my problem-solving abilities. The experience prepared me for future opportunities in AI.

Notes

What changed?

This was the baseline prompt with no prompt engineering techniques applied.

What improved?

Not applicable.

What still failed?

The response was generic and could describe almost any internship. It lacked context, structure, and personal experiences.

Version 1 : Role Assignment

Technique**Role Assignment****Prompt**

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Write a case study about my internship.

Representative Output

This internship allowed me to strengthen my technical knowledge while working on practical AI-related tasks. It highlights my learning process, the skills I developed, and my growth throughout the internship.

Notes**What changed?**

I assigned the AI the role of a portfolio writer.

What improved?

The writing sounded more professional and was better suited for a portfolio instead of a general summary.

What still failed?

The response still lacked information about my internship because no context had been provided.

Version 2 : Context and Motivation

Technique

Context and Motivation

Prompt

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Write a case study about my internship. I am completing the FlyRank AI Engineering Internship to build practical skills in Python, machine learning, and AI because I want to apply mathematics to real-world AI problems.

Representative Output

During the FlyRank AI Engineering Internship, I have been building practical skills in Python, machine learning, and AI while working with real datasets. My goal is to connect my mathematics background with practical AI applications through hands-on learning.

Notes

What changed?

I added context about my internship and explained why I joined it.

What improved?

The response became more specific to my learning journey and reflected my motivation instead of describing internships in general.

What still failed?

The response still lacked a clear structure and was presented as one long paragraph.

Version 3 : Few-Shot Examples

Technique

Few-Shot Examples

Prompt

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Write a case study about my internship. I am completing the FlyRank AI Engineering Internship to build practical skills in Python, machine learning, and AI because I want to apply mathematics to real-world AI problems.

Example

Problem: I wanted practical experience instead of only studying theory.

What I Did: I worked on real projects and documented my learning.

Outcome: I built confidence and developed practical skills.

Write my case study in a similar style using only my information.

Representative Output

Problem: I wanted to move beyond theory and understand how machine learning works in practice.

What I Did: I joined the FlyRank AI Engineering Internship and learned Python, AI, and machine learning through hands-on projects.

Outcome: I built a stronger foundation in AI and became more confident approaching unfamiliar technical concepts.

Notes

What changed?

I provided an example showing the style and format I wanted.

What improved?

The AI followed the desired style more closely, producing a more consistent and focused response.

What still failed?

The AI still made assumptions instead of asking for missing information.

Version 4 : Output Structure

Technique

Output Structure

Prompt

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Write a case study about my internship. I am completing the FlyRank AI Engineering Internship to build practical skills in Python, machine learning, and AI because I want to apply mathematics to real-world AI problems.

Organize the response using the following headings:

- The Problem
- What I Did and Decided
- What Came of It
- What I Would Do Differently

Representative Output

The Problem

I wanted to learn machine learning through practical experience instead of relying only on theory.

What I Did and Decided

I explored machine learning notebooks, learned AI workflows, and used AI tools to understand unfamiliar concepts step by step.

What Came of It

I developed a better understanding of machine learning notebooks and became more confident using AI to support my learning.

What I Would Do Differently

I would focus on learning one concept at a time instead of trying to understand everything immediately.

Notes

What changed?

I specified the output structure.

What improved?

The response became easier to read and followed a logical flow, making it more suitable for a portfolio.

What still failed?

The writing still sounded somewhat like standard AI-generated content.

Version 5 : Step Decomposition

Technique

Step Decomposition

Prompt

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Before writing anything, interview me one question at a time until you understand my project, the problem I wanted to solve, the decisions I made, the challenges I faced, and the outcome. After gathering enough information, write a portfolio-ready case study using the headings:

- The Problem
- What I Did and Decided
- What Came of It
- What I Would Do Differently

Use only the information I provide. Do not invent achievements or fill in missing details.

Representative Output

Instead of writing immediately, the AI first asked follow-up questions about my goals, decisions, challenges, and results. After collecting the information, it generated a portfolio case study that accurately reflected my own experience.

Notes

What changed?

I divided the task into two steps: gather information first, then write.

What improved?

The final output became much more accurate and personal because the AI collected the information it needed before writing. It relied less on assumptions and reflected my own experience.

What still failed?

The process took longer because it required an interactive conversation, but the improved quality made the extra time worthwhile.

Cross-Model Comparison

Criteria	Claude	ChatGPT
Tone	More conversational and reflective.	More direct and structured.
Accuracy	Stayed closer to the provided information and avoided unnecessary assumptions.	Followed instructions well but occasionally filled in missing details unless explicitly told not to.
Structure	Produced a natural narrative while maintaining the requested organization.	Followed the requested headings consistently and produced a well-organized response.
Failure Points	Sometimes asked broader follow-up questions than necessary before writing.	Occasionally assumed details that were not provided and required additional editing.

Final Reusable Prompt Template

You are a portfolio writer who helps students create clear and honest case studies for technical internships. Before writing, interview me one question at a time until you understand the project, the problem I wanted to solve, the decisions I made, the challenges I faced, and the outcome. Then write a portfolio-ready case study using the headings: The Problem, What I Did and Decided, What Came of It, and What I Would Do Differently. Use only the information I provide, avoid inventing details or achievements, and write in a direct, warm, plain, and specific style suitable for recruiters.

Reflection

This exercise showed me that adding more information to a prompt is not always enough. The biggest improvements came from providing meaningful context and breaking the task into clear steps before asking the AI to generate an answer. Comparing Claude and ChatGPT also showed that while both models followed the instructions well, they differed in tone and how they handled missing information. The final prompt is reusable because it focuses on the process of gathering information before writing, making it applicable to a wide range of portfolio projects.